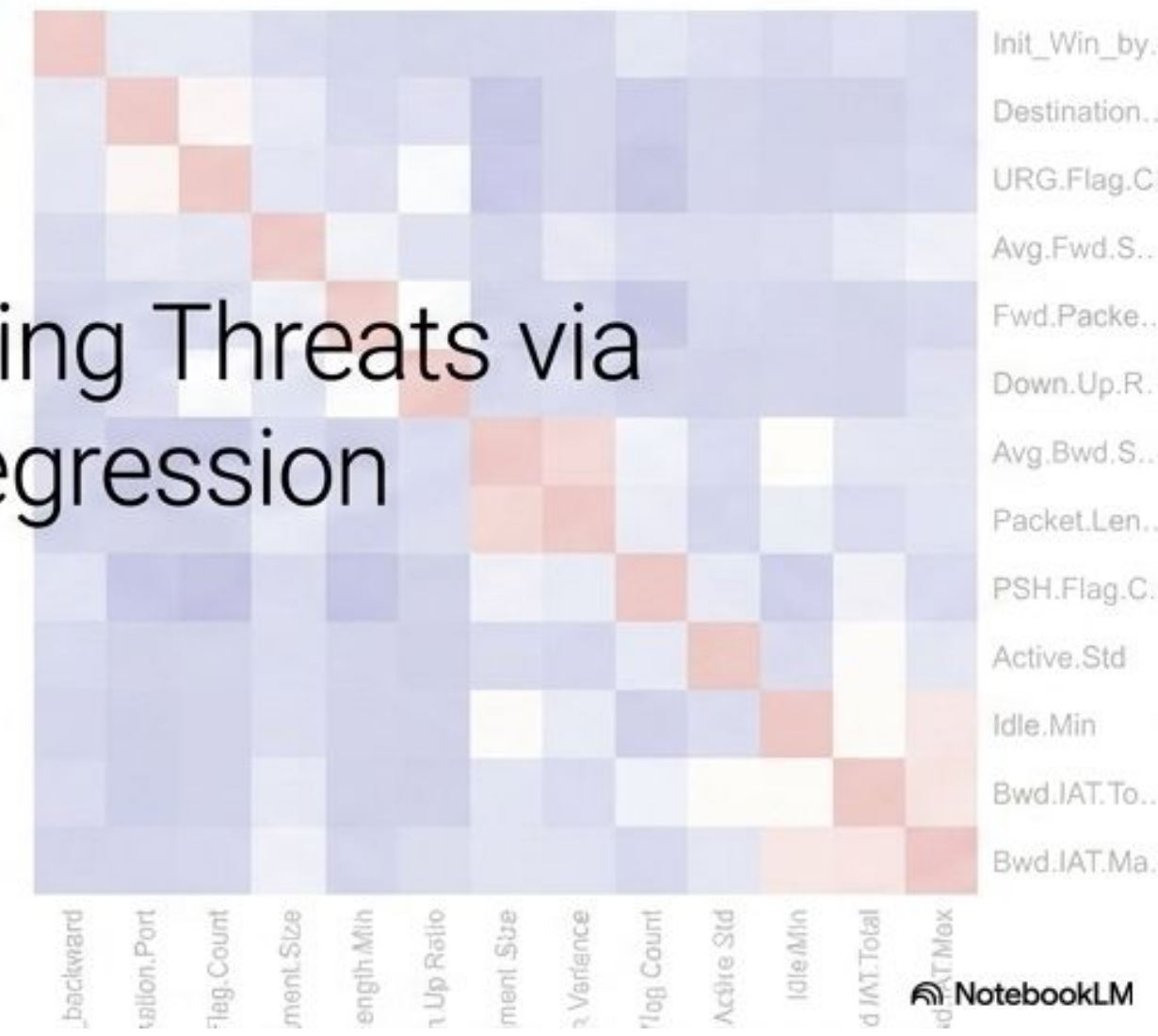


Statistical Analysis of CICIDS2017 Netousion Data Data

Mapping Normal Traffic and Isolating Threats via
Multiple Linear & Logistic Regression



Defining the Baseline and the Threat: Research Questions

DATASET TERRAIN

Dataset: CICIDS2017
(Canadian Institute for
Cybersecurity).

Scope: 79 columns describing
packet sizes, timing, TCP
flags, and routing.

Sample: ~250 rows sampled
per raw CSV file across 8 days
of traffic. Cleaned to remove
infinities and NAs.

Phase 1: Establish Normal (MLR)

Target: Benign traffic only.

Question: Which network
flow features best predict
the total byte volume
(Flow.Bytes)?

Goal: Build a statistical
baseline of normal traffic
shape.

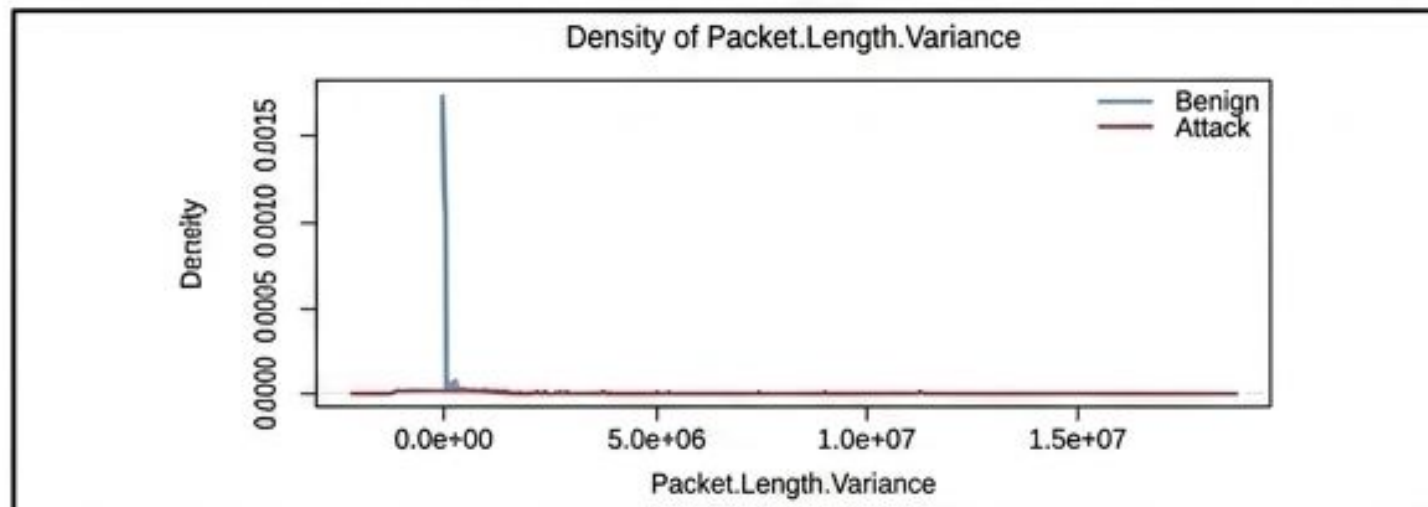
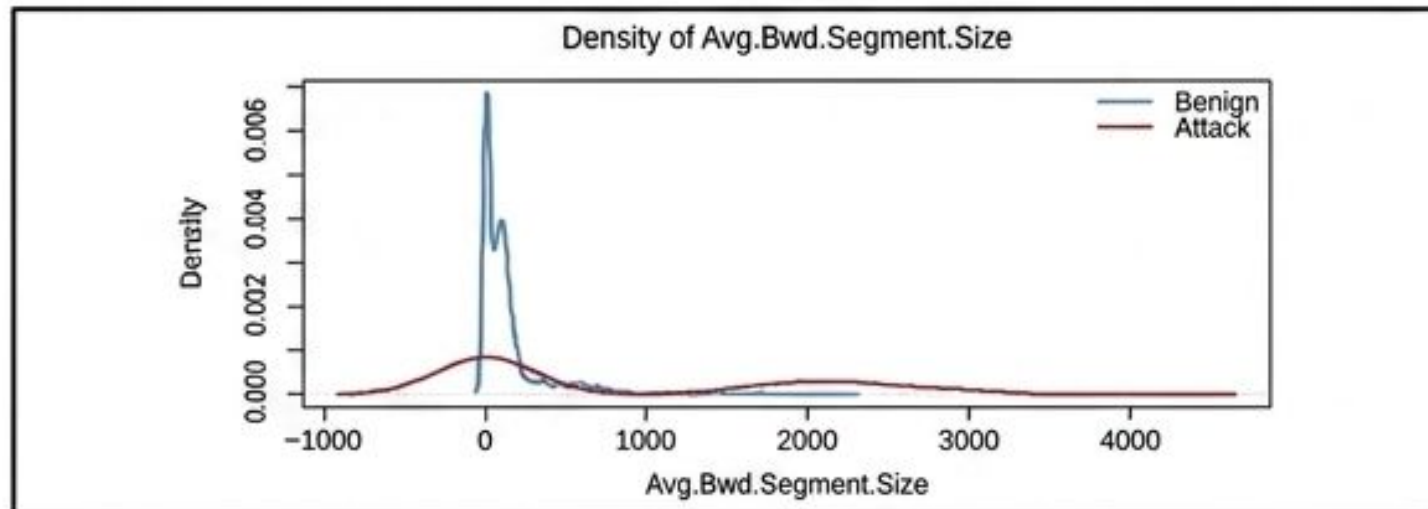
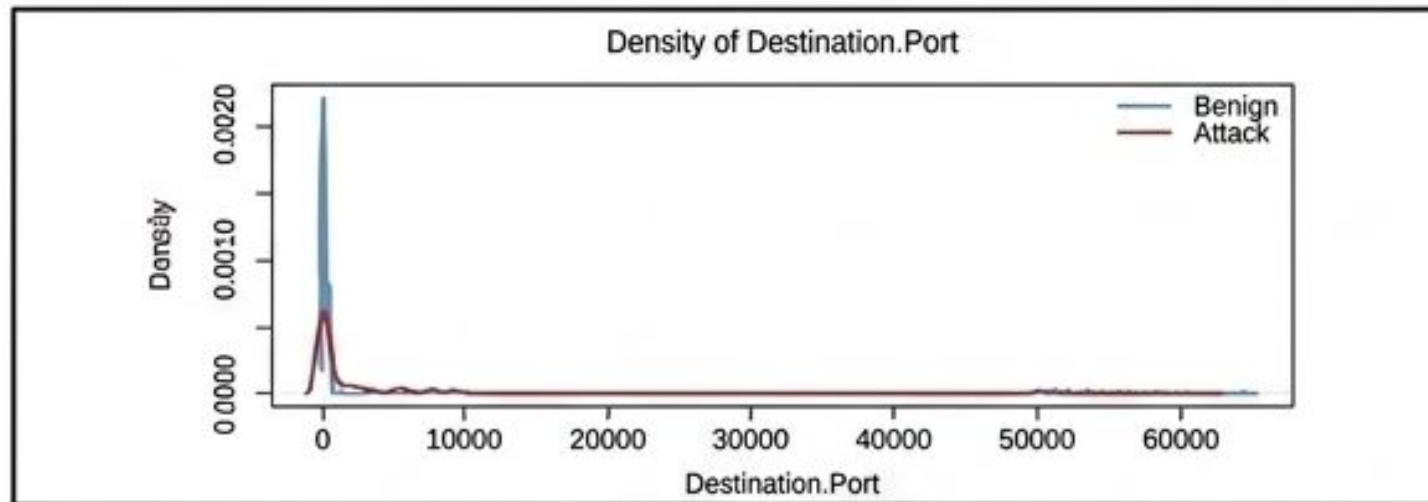
Phase 2: Catch the Culprit (Logistic)

Target: All traffic.

Question: Which features
distinguish attack flows
from benign flows?

Goal: Identify risk factors
and classify traffic.

Descriptive Profiles: Attack Traffic is Highly Dispersed

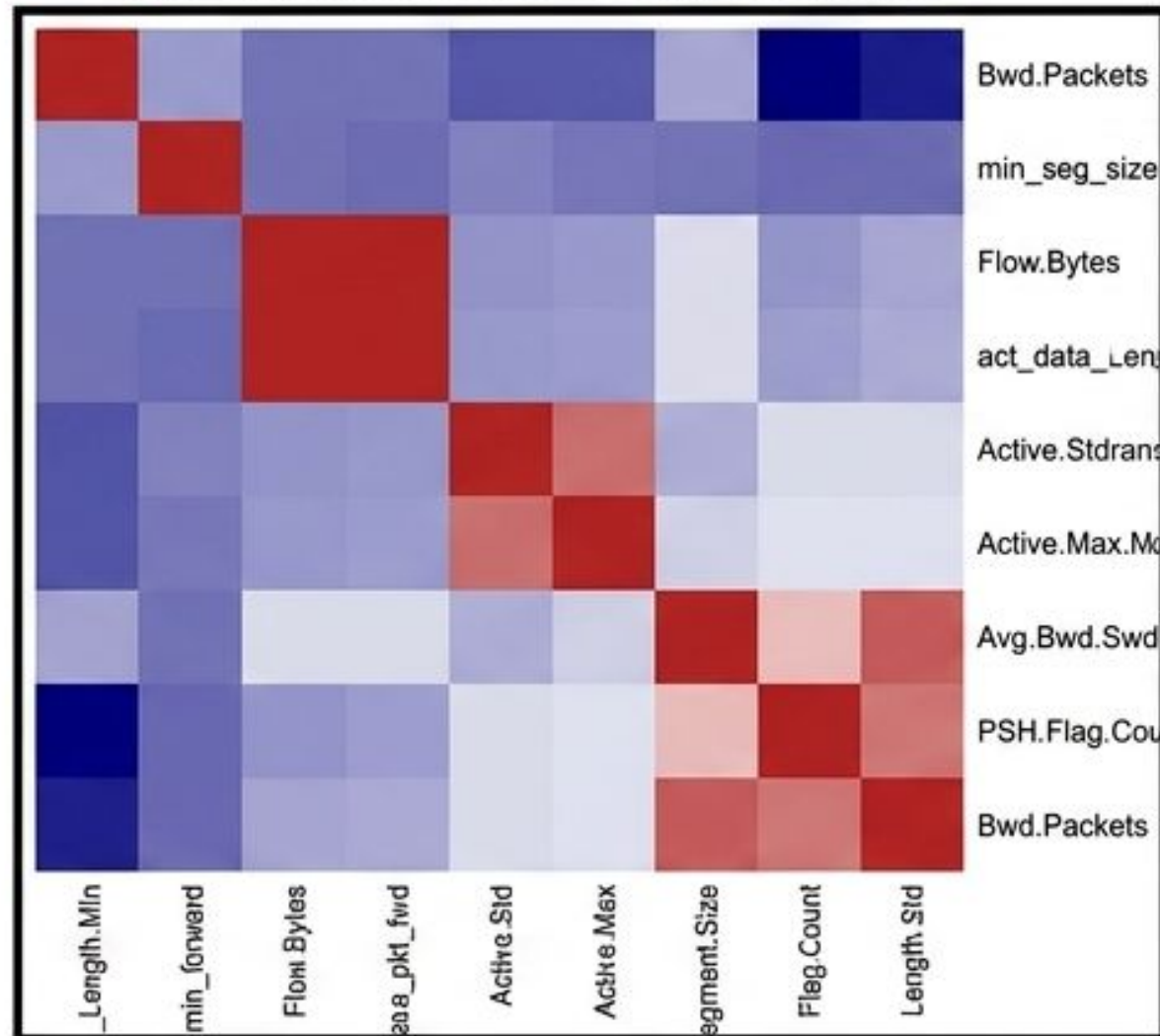


Visual Finding:
The blue lines (benign) show tight, predictable clustering. The red lines (attack) demonstrate massive variance, heavy tails, and unpredictable spread across key features.

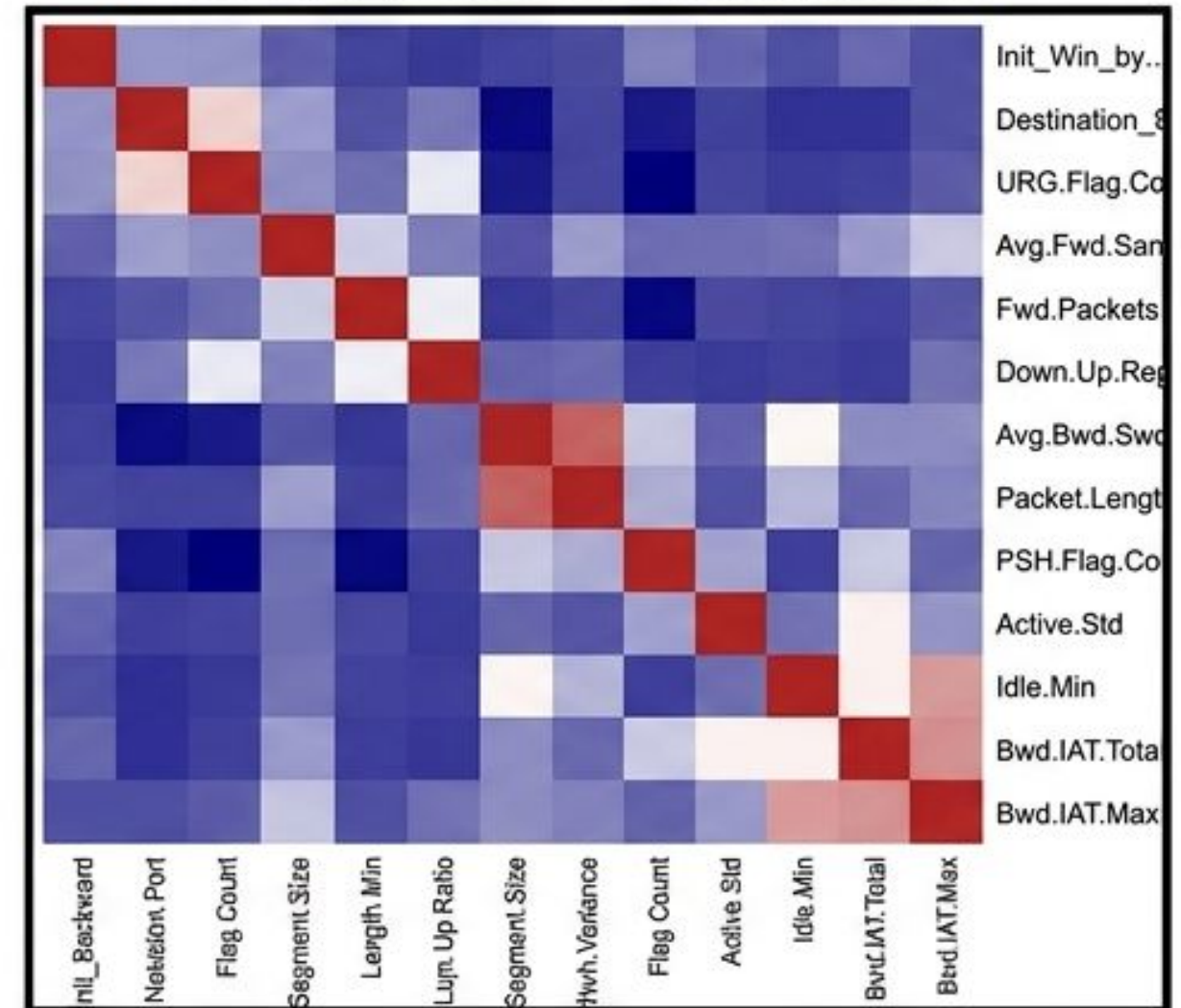
Key Insight:
Attackers must manipulate standard packet sizes and port usage to force entry or execute denial-of-service, leaving an observable statistical footprint.

Variable Correlations and Multicollinearity Pruning

MLR Predictor Correlations



GLM Predictor Correlations



Observation: Both heatmaps show intense clusters of correlation (dark navy and firebrick red), particularly among packet size and length variables.

Methodological Action: To ensure model stability, all highly correlated pairs ($|r| > 0.9$) were pruned prior to stepwise variable selection.

The Port Category Anomaly: Where Attacks Hide

Context: We mapped Destination.Port to IANA ranges: Well-known (0–1023), Registered (1024–49151), and Dynamic (49152–65535).

Row-wise Attack Rates	
1	Dynamic Ports: 95.8% Benign 4.2% Attack
2	Well-known Ports: 82.7% Benign 17.3% Attack
3	Registered Ports: 55.6% Benign 44.4% Attack

A Chi-square test confirms a massive, highly significant relationship between destination port category and attack status.

$\chi^2 = 178.69, df = 2, p < 2.22e-16$

MLR Setup: Modeling the Shape of Normal Traffic

Model Scope: First-order terms only (no interactions or polynomials).

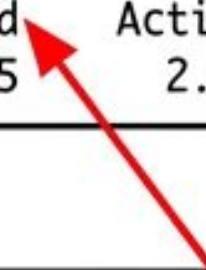
Variable Selection: Applied Stepwise BIC on the benign-only dataset.

Selected Predictors: 8 variables retained to predict Flow.Bytes.

Console

```
Multicollinearity (VIF)
vif_values <- car::vif(mlr_model)
print(vif_values)
```

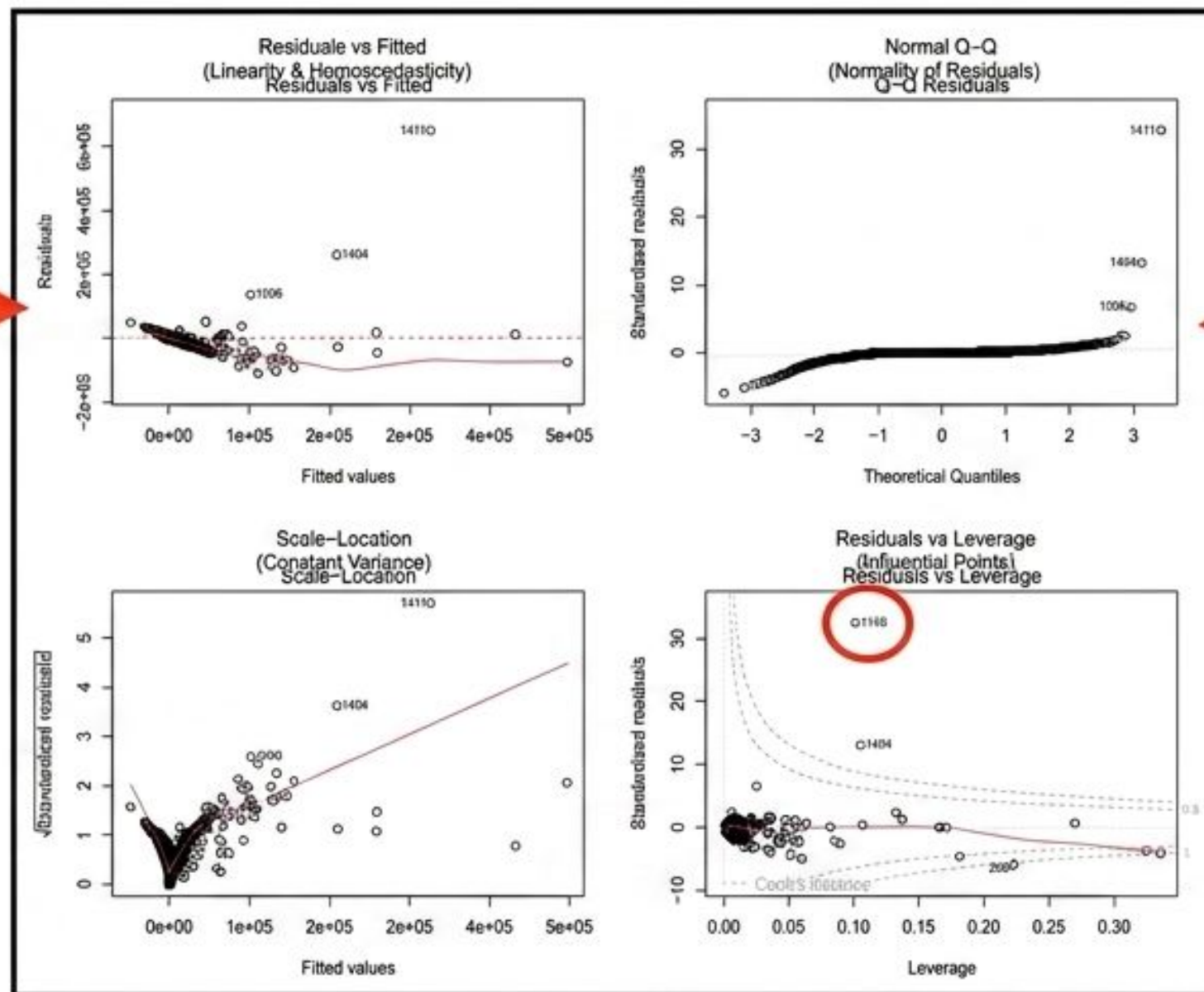
##	act_data_pkt_fwd	PSH.Flag.Count	Avg.Bwd.Segment.Size
##	1.720699	2.261688	5.937063
##	Bwd.Packet.Length.Std	Bwd.Packet.Length.Min	min_seg_size_forward
##	6.830515	1.769534	1.038479
##	Active.Std	Active.Max	
##	2.406375	2.545160	



Multicollinearity Checked: Max VIF is 6.83, well below the standard threshold of 10. The model predictors are stable.

MLR Diagnostics: Identifying Systemic Assumption Violations

Heteroscedasticity:
The Residuals vs Fitted plot shows a distinct funnel shape. Validated by NCV Test ($p < 2.22e-16$).

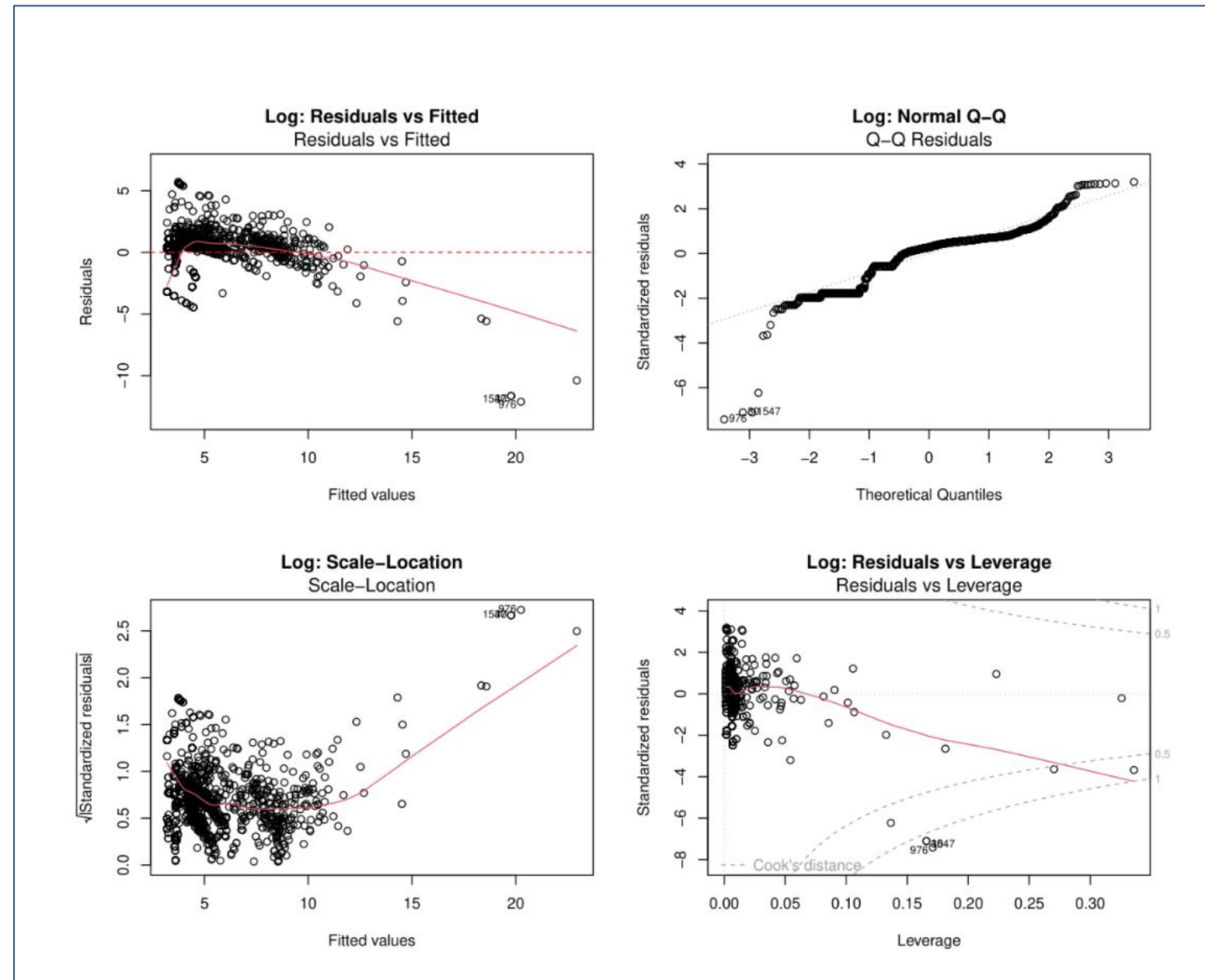


Non-Normality: The Q-Q plot shows massive tail deviations. Validated by Shapiro-Wilk ($p < 2.2e-16$).

The Culprit (Influential Outlier): Row 1198 has an extreme leverage of 0.99, artifactually inflating the original model's R-squared to 0.99.

Enhancing the MLR Model: Transformation and Pruning

1. **Response Transformation:** Applied a log-transformation **log1p(Flow.Bytes)** to compress the scale and stabilize variance.
2. **Outlier Removal:** Pruned the single extreme leverage point (Row 1198) that was warping the regression plane.



Synthesis: The Scale-Location plot now shows a drastically flattened, stabilized line. The Q-Q plot is significantly tighter. While network traffic is inherently noisy, the transformation allows for reliable inference.

MLR Interpretation: The Drivers of Benign Traffic Volume

Quality of Fit: R-squared = 0.6249 (A realistic fit after removing the leverage outlier).
Overall Model Significance: $F(8, 1601) = 333.35$, $p < 2.2e-16$. All 8 predictors remain statistically significant.

Broad Strokes Interpretation

Impact per 1-Standard Deviation Change:

- Dominant Driver 1: Avg.Bwd.Segment.Size (Accounts for 82.99% of Response SD)
- Dominant Driver 2: act_data_pkt_fwd (Accounts for 45.00% of Response SD)

Confidence Intervals

```
confint(mlr_model, level = 0.95)
```

##	2.5 %	97.5 %
## (Intercept)	-1.300650e+04	-3.892136e+03
## act_data_pkt_fwd	1.671243e+03	1.991878e+03
## PSH.Flag.Count	-9.793332e+03	-2.656970e+03
## Avg.Bwd.Segment.Size	9.490163e+01	1.132506e+02
## Bwd.Packet.Length.Std	-8.363792e+01	-6.382593e+01
## Bwd.Packet.Length.Std	-1.183792e+01	-8.382593e+01
## Bwd.Packet.Length.Min	-1.183284e+02	-8.754250e+01
## min_seg_size_forward	9.615739e+01	4.342465e+02
## Active.Std	-4.758192e-02	-2.940258e-02
## Active.Max	1.057234e-02	1.546501e-02

Logistic Regression Setup: Classifying Threats

Target: Predicting Attack_Flag (Binary: 0 = Benign, 1 = Attack). Overall attack rate in data is 19.3%.

Data Split: 70% Training (1,397 rows) / 30% Testing (6100 rows).

Model Scope: First-order terms only. Iterative Stepwise BIC and VIF-based pruning selected 11 final predictors. Max VIF: 4.37.

Diagnostic Alert: Active.Std was flagged for partial separation (Standard Error > 10). The variable was retained for predictive accuracy, but its odds ratio is mathematically unstable and will not be interpreted.

Logistic Model Performance & Predictive Power

The overall model is **highly significant**. Likelihood Ratio Test $X^2 = 878.25$, $p = 2.94e-181$.

Accuracy: 91.2%

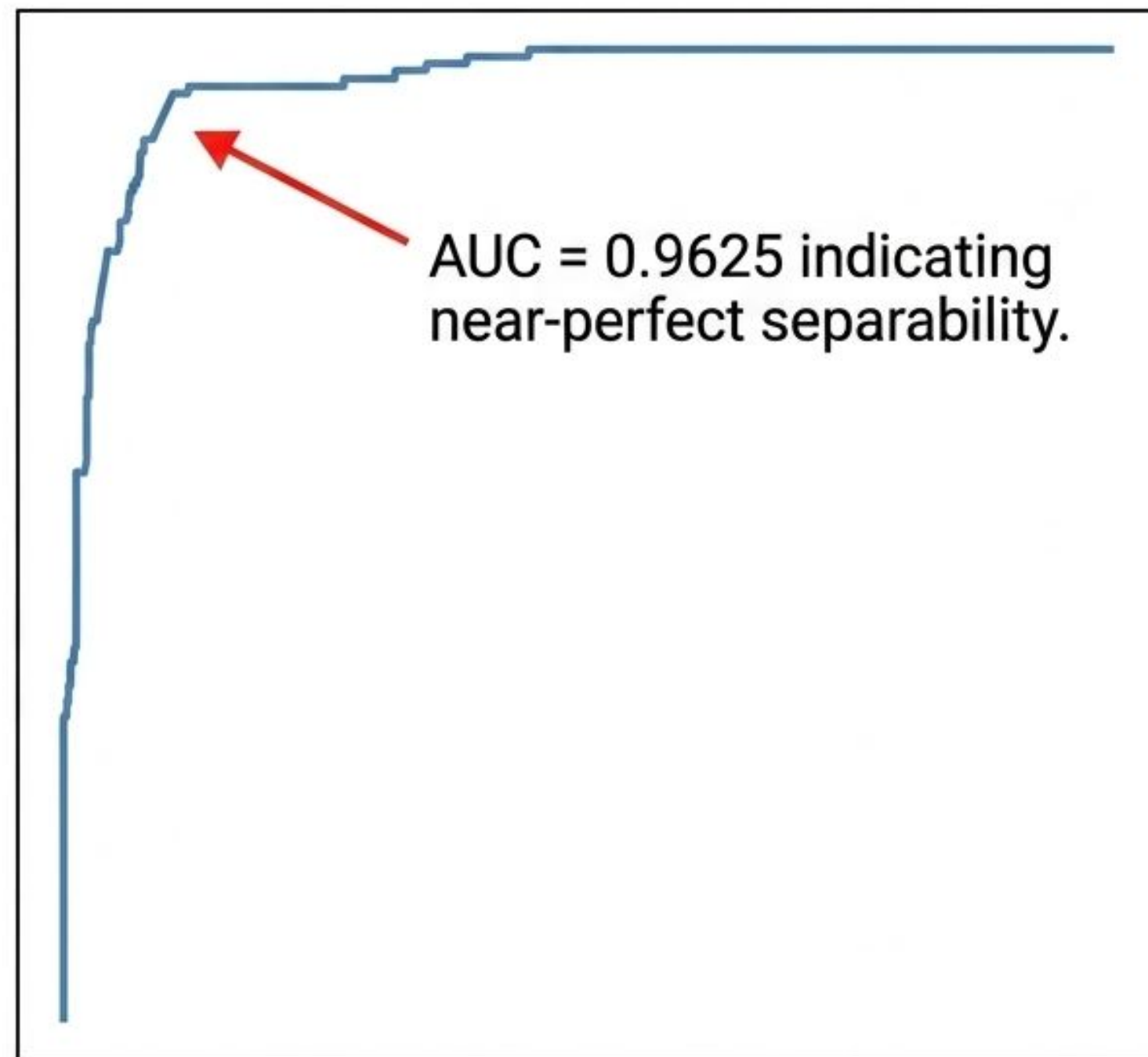
(at 0.5 probability cutoff)

Sensitivity: 95.3%

(Model successfully catches 95.3% of true attacks)

Specificity: 89.0%

(Using Youden's Index optimization)



Broad Strokes Interpretation: The Anatomy of an Attack

To interpret slopes practically, we measured the percentage change in the odds of an attack per 1-Standard Deviation increase in each predictor.

High-Risk Signals (Increases Odds)

Idle.Min: **LARGE** effect (+319.3% increase in odds per 1-SD).

Down.Up.Ratio: **LARGE** effect (+187.8% increase in odds per 1-SD).

Interpretation: Attacks often feature prolonged idle states and asymmetrical upload/download behavior.

Protective Signals (Decreases Odds)

Fwd.Packet.Length.Min: **LARGE** negative effect (-100% decrease).

Avg.Fwd.Segment.Size: **LARGE** negative effect (-100% decrease).

Interpretation: Normal forward packet sizes and minimum lengths strongly suggest benign, standard communication.

Summary Insights: Defining Normal to Isolate the Abnormal

Phase 1: The Baseline (MLR)	Phase 2: The Anomaly (Logistic)
Normal traffic volume is highly predictable and driven primarily by base segment sizes (Avg.Bwd.Segment.Size) and active data packets. Systemic Trait: The data is inherently heavy-tailed and heteroscedastic, requiring log-transformations for baseline inference.	Malicious traffic deviates sharply from this baseline. It operates on Registered Ports (44.4% attack rate) and manifests via anomalous timing (Idle.Min) and asymmetrical flow (Down.Up.Ratio). Systemic Trait: These deviations allow a 1st-order logistic model to isolate attacks with 91.2% accuracy and an AUC of 0.9625.

By mathematically defining the shape of benign flow, anomalous network intrusions become statistically glaring, enabling highly sensitive (95.3%) automated detection.